

NATITINGOU, Benin

WMO: 653190

Lat: 10.317N

Lon: 1.383E

Elev: 1512

StdP: 13.91

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	61.6	64.1	25.6	20.7	78.8	28.6	23.9	79.3	13.7	79.8	12.1	80.1	0.8	140	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	21.2	100.0	72.3	98.3	73.2	96.9	73.2	83.3	92.6	81.9	91.3	80.8	90.1	5.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	170.1	90.3	79.5	162.1	88.7	78.3	155.5	87.2	48.8	92.8	47.2	91.5	45.8	90.3	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.3	9.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.1	79.3	83.5	87.0	85.9	83.3	80.5	78.2	77.6	78.6	80.3	80.2	79.5
	DBStd	3.86	3.22	3.20	2.44	2.90	2.77	2.58	2.08	1.95	2.01	1.85	1.78	2.86
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11365	910	937	1146	1077	1033	915	874	854	857	940	907	915
	CDD65	5890	445	517	681	627	568	465	409	389	407	475	457	450
	CDH74	63966	5035	6723	9616	8356	6871	4727	3083	2587	2930	4222	4879	4937
	CDH80	30326	2570	3826	5628	4483	3196	1753	781	532	835	1695	2495	2532
Wind	WSAvg	3.3	3.7	3.3	3.6	4.2	4.0	3.5	3.6	3.2	2.3	2.4	2.6	3.1
Precipitation	PrecAvg	49.1	0.0	0.2	0.9	3.2	4.7	5.9	8.3	10.8	10.0	4.1	0.7	0.1
	PrecMax	69.6	1.7	1.3	2.5	10.4	8.0	10.7	17.2	17.8	19.0	9.6	8.1	2.4
	PrecMin	32.7	0.0	0.0	0.0	0.7	1.5	0.8	2.8	3.1	4.4	0.6	0.0	0.0
	PrecStd	8.5	0.3	0.3	0.7	1.9	1.6	2.2	3.2	3.2	3.0	2.0	1.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.6	101.2	102.5	101.3	97.1	93.2	88.9	88.0	89.8	94.3	97.4	97.0
		MCWB	68.3	71.2	71.6	74.1	80.7	79.0	78.4	78.8	78.9	76.3	68.4	67.7
	2%	DB	95.7	98.7	100.5	99.0	94.6	90.9	87.1	85.9	88.0	91.7	95.4	95.2
		MCWB	67.9	71.9	71.6	75.3	79.7	78.8	77.8	77.7	78.6	77.0	69.3	66.6
	5%	DB	93.6	96.9	98.8	96.9	93.0	89.4	85.8	84.4	86.3	89.9	93.5	93.6
		MCWB	67.6	70.9	73.0	77.0	79.2	78.3	77.3	77.0	78.2	77.8	69.4	66.3
	10%	DB	91.4	95.1	97.2	94.8	91.3	87.8	84.3	82.9	84.8	88.1	91.5	91.5
		MCWB	66.8	69.7	73.7	78.1	79.1	78.0	76.7	76.4	77.6	77.4	69.8	66.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.9	81.8	83.2	85.2	86.0	83.5	80.5	80.3	81.2	82.6	79.5	80.0
		MCDB	91.1	96.1	96.2	95.2	94.3	89.5	86.3	86.2	87.2	90.0	89.3	90.9
	2%	WB	74.3	78.6	81.6	83.0	83.3	81.2	79.2	78.9	80.0	80.9	77.9	76.5
		MCDB	90.4	93.8	95.1	93.0	91.3	88.1	85.3	84.6	86.0	88.7	87.7	88.6
	5%	WB	71.9	75.5	80.0	81.3	81.6	79.9	78.1	77.8	78.9	79.5	76.5	73.2
		MCDB	89.5	91.1	93.5	91.8	90.0	87.1	84.2	83.3	85.2	87.4	86.6	86.9
	10%	WB	69.4	73.5	78.5	79.8	80.1	78.7	77.1	76.8	77.9	78.3	74.9	70.3
		MCDB	88.0	89.9	92.0	90.8	88.9	85.8	83.2	82.1	83.8	86.1	84.8	85.5
Mean Daily Temperature Range	5% DB	MDBR	25.1	24.0	21.2	17.6	15.3	13.7	11.7	11.1	13.6	17.8	25.8	27.0
		MCDBR	28.8	26.2	24.3	20.6	17.8	15.8	14.1	13.4	15.8	20.0	28.3	28.9
		MCWBR	14.9	13.1	11.1	8.0	7.5	6.8	6.5	6.7	8.2	9.0	11.1	13.4
	5% WB	MCDBR	27.7	23.2	19.6	17.3	16.0	14.8	13.4	12.8	15.2	18.3	23.0	27.2
		MCWBR	17.5	13.8	9.9	8.6	9.2	8.1	7.1	7.2	8.6	9.7	11.4	16.6
Clear-Sky Solar Irradiance	taub	0.658	0.723	0.797	0.710	0.618	0.567	0.518	0.504	0.510	0.596	0.576	0.575	
	taud	1.658	1.547	1.455	1.657	1.806	1.939	2.117	2.158	2.113	1.854	1.829	1.817	
	Ebn at Noon	214	207	195	211	227	235	249	256	257	233	233	231	
	Edh at Noon	79	91	101	82	69	60	50	49	52	66	66	66	
All-Sky Solar Radiation	RadAvg	1714	1780	1853	1874	1818	1681	1496	1412	1570	1776	1787	1705	
	RadStd	63	62	73	58	64	51	52	69	66	67	46	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page